

# Seyedeh Zahra (Mahya) Hamed

 (+98) 917-225-8973

 mahyahamedi98@gmail.com

 [Portfolio](#)

 [Mahya Hamed](#)

## Education

**MSc. Medical Informatics** - GPA: 3.61/4

01 Oct 2020 - 20 Dec 2023

Shahid Beheshti University of Medical Sciences - Fully Scholarship

Thesis: Application of Machine Learning in Breast Cancer Survival Prediction: a Multimethod and Multicenter Comparison

- Web and Windows programming: 18.25/20
- Neural networks and intelligent medical systems: 19.5/20
- Thesis: 19/20
- Management in Medical Informatics: 18.5/20

**BSc. Health Information Technology** - GPA: 3.55/4

15 Sep 2016 - 22 Aug 2020

Shiraz University of Medical Sciences - Fully Scholarship

Thesis: investigating the level of compliance with the requirements in selected virtual assistants in the field of mental health

- General Mathematics: 20/20
- Research Project: 19/20
- Computer network and systems security: 18.9/20
- Health Informatics: 18.5/20
- Research Project: 19/20
- Health Information Management: 18/20

## Research Interests

- **Innovation:** AI, Machine learning, Precision Medicine, Neuroscience, LLMs, eHealth, IoT, Gamification
- **Design and Evaluation:** Digital Health Product Design, Strategic Management

## Skills

- **Programming:** Python (Web scraping - Sklearn - TensorFlow - Scikit-learn - Tesseract - Streamlit), HTML, CSS
- **Research Skills:** Survey Design, Data Analysis, Machine Learning, SPSS, Data Gathering, SPSS, Interviewing
- **Product Design:** Benchmark, Need Assessment, User Research, Figma, Adobe photoshop
- **Other Technical Skills:** MySQL, Software Testing, Postman, AI Agent Development, Fine Tuning, RAG
- **Soft Skill:** Team Working, Speech and Presentation, Project Management, Problem Solving, Stress Management
- **Language Skills:** Persian (Native), English (IELTS 7), Spanish (Beginner)

## Research & Publications

- **Seyedeh Zahra Hamed, et al “Design and Evaluation of Gamification Approaches in a Personalized Health Mobile Application for Lifestyle Optimization”**  
2026, Not published yet
- **Sohrab Almasai, Reza Rabiei ,Seyedeh Zahra Hamed, et al “Healthcare Dashboards: An Umbrella Review of Applications, Benefits, Design and Challenges”**  
Digital Health Journal, 2026 - [DOI](#)
- **Seyedeh Zahra Hamed, Hassan Emami, Maryam Khayamzadeh, Mehrad Aria, Reza Rabiei , et al “Application of Machine Learning in Breast Cancer Survival Prediction Using a Multimethod Approach”**  
Published by Nature in Scientific Reports Journal, 2024 - [DOI](#)
- **Seyedeh Zahra Hamed, Hassan Emami, Maryam Khayamzadeh, Mehrad Aria, Reza Rabiei , et al. “Providing an Efficient Model Based on Deep Learning to Predict 5-year Survival in Women with Breast Cancer”**  
Accepted at the 17th International Conference on Health Informatics, 2024
- **Mehrdad Karajizadeh, Reza Nikandish, Omid Yousefianzadeh, Zahra Hamed, and Hamid Reza Saaidnia. “Usability Evaluation of the Electronic Medical Record of the Rapid Response Team: a Case Study.”**  
Published by Applied Health Information Technology Journal, 2021 - [DOI](#)

## Work Experience

### Digital Product Manager LifeandMe | Tehran - Iran

01 Sep 2022 - Present

A DTC genetic testing company

- Doing user research and managing the design and development of an Android application for our customers, providing health reports based on genetic and epigenetic data analysis.
- Exploring Interoperability Standards and nomenclatures to design genetic and epigenetic databases and the data analysis process automation.
- Using fine-tuning and retrieval-augmented generation (RAG) in large language models (LLMs) to provide more comprehensive recommendations and develop a chatbot for our users.
- Led the design and development of a web-based panel to manage internal workflows, such as logistics, financial operations, and CRM.
- Developed a WhatsApp automation tool and Telegram bot using Python, which was utilized as a marketing and CRM tool.

### Research Assistant Multi-Center | Shiraz & Tehran - Iran

04 Jan 2022 - 20 Dec 2023

- Interviewed Patients and care providers for data gathering, cleaning data and entering them into EHR.
- Machine Learning and Deep Learning Modeling

### Digital Product Manager YouMate | Tehran - Iran

01 Apr 2021 - 31 Oct 2022

- Managed the design and development of a web-based panel to analyze medical lab tests using OCR.
- Developed the OCR functionality using Python (PaddleOCR, Tesseract).
- Participated in creating a medical lab dictionary to train the OCR
- Participated in Producing Multimedia educational content for medical lab factors introduction and lifestyle modification.

### Product Designer YouMate | Tehran - Iran

01 Sep 2020 - 31 Mar 2021

- Designed a mobile application MVP in the field of health.
- Participated in The Design and development of a personalized PDF named CoWin that provides personalized instructions in the fields of nutrition, exercise, mindful eating, and mental health to prevent coronavirus.

## Other Activities

**Volunteer Work** - Managing the development of a self-care application for cystic fibrosis patients and a specialized EHR system for CF healthcare centers in Iran, under the supervision of the Iran CF Foundation.

2023-Present

**Collaboration with The Academy of Medical Sciences** - Participated with the Digital Health Committee in creating an infographic document on elevating digital health literacy

July 2024

**Psychology based HR Platform** - Managing the design and development of a human resource management platform based on psychological tests that were used by a psychiatric resident in his thesis.

Aug 2022 - Feb 2023

**Personalized Medicine National Document** - Member of a multidisciplinary group of writers

Nov 2021-May 2022

The goal of this document was to provide guidance for enhancing PM in Iran by drawing inspiration from successful cases around the world.

## Selected Honors & Awards

**2023- YouMate** was selected as one of the top teams in Iran's Intelligent Leap in Digital Economy Plan.

This was because of a web-based panel I managed, designed to analyze medical lab tests using OCR.

**2020- 3rd** Out of all applicants in the Nationwide Universities Entrance Exam for Master of Medical Informatics

**2020- 5th** Out of all applicants in the Nationwide Universities Entrance Exam for Master of Health IT

**2016- Top 2%**, Out of 500,000 applicants in the Nationwide Universities Entrance Exam

## Selected Courses

**2025-** Introduction to Retrieval Augmented Generation (RAG) - Coursera Instructor Network

**2024-** User Research of Digital Products - Userx Academy

**2024-** Machine Learning Foundations for Product Managers - Duke University [Certificate](#)

**2023-** Digital Product Management: Modern Fundamentals - University of Virginia [Certificate](#)

**2023-** Data Science in Stratified Healthcare and Precision Medicine - The University of Edinburgh [Certificate](#)

**2020-** Advanced programming using Python - Maktabkhone [Certificate](#)

**2020-** Introduction to programming using Python - Maktabkhone [Certificate](#)

**2019-** Bioinformatics Summer School - Shiraz University of Medical Sciences [Certificate](#)